

ZIZHOU (GARY) LIU

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EDUCATION

Columbia University

M.S. in Computer Science, Natural Language Processing Track

Feb 2023

New York, NY

Columbia University

B.S. in Computer Science, Intelligent Systems Track

May 2021

New York, NY

TECHNICAL SKILLS

Languages & Frameworks: Python, PyTorch, Hugging Face Transformers, TensorFlow, LangChain, Java, C, Git

ML & LLM Systems: Fine-tuning, distillation, quantization, inference optimization, evaluation, RAG, agentic workflows/harness, ranking, recommendation, entity resolution

Data & Infrastructure: BigQuery, Pinecone, vector search, large-scale labeling, production model pipelines

PROFESSIONAL EXPERIENCE

Loxo Talent Intelligence Platform

Machine Learning Tech Lead

Jun 2026 – Present

- Lead the ML team responsible for Loxo's agentic recruiting platform, spanning candidate discovery, longlisting agent, shortlisting agent, ranking, and personalized outreach agent.
- Define the architecture and roadmap for coordinated recruiting agents powered primarily by specialized 0.6B–4B language models and frontier LLMs, balancing model quality, latency, cost, and production reliability.
- Drive end-to-end execution across data strategy, model training, evaluation, inference optimization, deployment, and cross-functional integration with product, ML, and data teams.

Machine Learning Engineer

Apr 2024 – May 2026

- Developed and optimized model pipelines for profile and experience merging, title resolution, company industry classification, resume matching, candidate ranking, and automated email outreach.
- Built high-scale talent intelligence systems achieving approximately 0.99 precision for profile and experience merging and supporting industry classification across 36.5 million companies.
- Fine-tuned and accelerated Qwen, Llama, Phi, and BERT-family models ranging from compact encoders to 13B parameters using task-specific training, distillation, quantization, and inference optimization.

Columbia University – Conversational AI Lab

Staff Research Associate, Full-time (Prof. Zhou Yu)

Oct 2023 – Apr 2024

Remote

- Developed production-oriented conversational AI systems for sales and education using LLMs, dialogue-flow orchestration, intent characterization, tree-of-thought reasoning, retrieval-augmented generation, BigQuery, and Pinecone.
- Built and deployed partner-facing chatbot solutions, translating research prototypes into reliable end-user workflows for organizations including Articulate AI and Richtech Robotics.

Black Sesame Technologies

Machine Learning Engineer

Jan 2023 – Oct 2023

San Jose, CA

- Improved DNN inference latency and power consumption by 2–3× through magnitude pruning, dynamic-step pruning, INT4/INT8 quantization, distillation, and equivalent layer, node, and model substitutions.
- Optimized production vision models (YOLO, CenterPoint, BiSeNet, etc.) for resource-constrained automotive hardware.

Symanto – Psychology AI

AI Research Intern

Jun 2020 – Aug 2020

Remote

- Developed a model-quality platform for evaluating, stress-testing, diagnosing, and retraining user-provided NLP models.

SELECTED RESEARCH EXPERIENCE

Columbia University – Speech Lab / DARPA SemaFor

Research Associate, advised by Prof. Julia Hirschberg

May 2022 – Aug 2025

New York, NY / Remote

- *Psychology-Grounded LLMs*: Led an interdisciplinary review of how psychological theories are incorporated across LLM training, evaluation, and application; identified conceptual gaps and proposed directions for theory-grounded research. Published as a first-author long paper at EACL 2026.
- *Social-Engineering and Malicious-Intent Detection*: Developed ConvoSentinel, combining message-level classification, conversation context, and retrieval-augmented generation to detect social-engineering attacks; improved F1 by 12% over baseline systems.
- *Controllable Generation for Machine Reading Comprehension*: Developed Question-Attended Span Extraction (QASE), a lightweight module for conditioning generative models on question-relevant evidence across Llama 2, Alpaca, and Flan-T5.
- Contributed to research on propaganda analysis, crisis narratives, misinformation detection, and psychologically informed NLP, resulting in publications at EACL, EMNLP, COLING, ICWSM, and related venues.

Columbia University – Social Media and Public Health

Research Assistant, advised by Prof. Luis Gravano and Prof. Daniel Hsu

Aug 2020 – Aug 2022

New York, NY

- *Prompt-Based Few-Shot Learning*: Studied BERT and RoBERTa, analyzing how verbalizer selection affects classification performance in zero-shot and few-shot settings.
- *Soft Verbalizers*: Proposed an approach that learns task-specific representations from multiple verbalizer embeddings, improving classification accuracy over manually selected verbalizers.

SELECTED PUBLICATIONS

Liu, Zizhou, Gong, Z., Ai, L., Hui, Z., Chen, R., Leach, C. W., Greene, M. R., Hirschberg, J. [A Review of Incorporating Psychological Theories in LLMs](#). **EACL 2026**.

Liu, J., Ai, L., **Liu, Zizhou**, Karisani, P., Hui, Z., Fung, M., Nakov, P., Hirschberg, J., Ji, H. [PropaInsight: Toward Deeper Understanding of Propaganda in Terms of Techniques, Appeals, and Intent](#). **COLING 2025**.

Ai, L., Kumarage, T., Bhattacharjee, A., **Liu, Zizhou**, Hui, Z., Davinroy, M., Cook, J., Cassani, L., Trapeznikov, K., Kirchner, M., Basharat, A., Hoogs, A., Garland, J., Liu, H., Hirschberg, J. [Defending Against Social Engineering Attacks in the Age of LLMs](#). **EMNLP 2024**.

Ai, L., Hui, Z., **Liu, Zizhou**, Hirschberg, J. [Enhancing Pre-Trained Generative Language Models with Question Attended Span Extraction on Machine Reading Comprehension](#). **EMNLP 2024**.

Ai, L., Gupta, S., Oak, S., Hui, Z., **Liu, Zizhou**, Hirschberg, J. [TweetIntent@Crisis: A Dataset Revealing Narratives of Both Sides in the Russia-Ukraine Crisis](#). **ICWSM 2024**.

Ai, L., **Liu, Zizhou**, Hirschberg, J. [Combating the COVID-19 Infodemic: Untrustworthy Tweet Classification using Heterogeneous Graph Transformer](#). **Mediate @ ICWSM 2023**.

Cao, I.*, **Liu, Zizhou**.*, Karamanolakis, G., Hsu, D., Gravano, L. [Quantifying the Effects of COVID-19 on Restaurant Reviews](#). **SocialNLP @ NAACL 2021**.